Alertreck: Offline Edge AI for Anti-Poaching Acoustic Surveillance:

Benchmarking Pretrained Embeddings, Metric Learning, Supervised Classification, and Anomaly Detection on Identical Field Hardware

BSc in Software Engineering (Machine Learning)

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DECLARATION

I, Orpheus Mhizha Manga, declare that this project proposal is my original work and has not been submitted for academic credit at this or any other institution. All sources of information consulted during its preparation have been appropriately acknowledged and cited in accordance with applicable academic referencing standards. The system design, comparative experimental framework, and supporting documentation are the result of my own research and engineering effort under the supervision of Hubert Apana at the African Leadership University.

Signature: Orpheus Mhizha Manga

Date: May 2026

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LIST OF ACRONYMS AND ABBREVIATIONS

AI	Artificial Intelligence
AST	Audio Spectrogram Transformer
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CNN	Convolutional Neural Network
Conv-AE	Convolutional Autoencoder
DCASE	Detection and Classification of Acoustic Scenes and Events
DDR	Design and Development Research
DOA	Direction of Arrival
DSP	Digital Signal Processing
Edge AI	Artificial Intelligence deployed on local edge computing devices
ESC-50	Environmental Sound Classification Dataset (50 classes)
GPS	Generalised Cross-Correlation with Phase Transform
GSM	Global System for Mobile Communications
HMM	Hidden Markov Model
INT8	8-bit Integer (quantization precision)
IoT	Internet of Things
GPS fix time	GPS fix accuracy
MFCC	Mel-Frequency Cepstral Coefficients
ML	Machine Learning
OC-SVM	One-Class Support Vector Machine
PAM	Passive Acoustic Monitoring
PAWS	Protection Assistant for Wildlife Security
ProtoNet	Prototypical Network
RAM	Random Access Memory

RFCx	Rainforest Connection
SDG	Sustainable Development Goal
SMS	Short Message Service
SNR	Signal-to-Noise Ratio
SVM	Support Vector Machine
GPS NMEA fix	Time Difference of Arrival
TFLite	TensorFlow Lite
UAV	Unmanned Aerial Vehicle
UNODC	United Nations Office on Drugs and Crime
VAD	Voice Activity Detection
XAI	Explainable Artificial Intelligence

ABSTRACT

Wildlife poaching across African savannah ecosystems continues to devastate biodiversity, with between 10,000 and 15,000 elephants killed annually and ranger coverage standing at just one per 72 km² far below what effective surveillance requires (Waldron et al., 2022; Elephanatics Foundation, 2024). Existing solutions fail in the field. Rainforest Connection requires a 50 Kbps continuous uplink, rendering it non-functional across most remote African parks (Lynam et al., 2025), while AudioMoth operates offline but delivers no real-time alerts, leaving threats unresponded to until physical device retrieval. This project designs, develops, and evaluates **Alertreck**: a fully offline edge AI acoustic monitoring system running on a Raspberry Pi 4 Model B (2 GB RAM) with a single USB microphone and a SIM808 GSM/GPS module, built specifically for remote conservation environments with no internet access.

The research conducts a formal four-paradigm comparative study currently unaddressed in the anti-poaching acoustic literature, benchmarking supervised multi-class classification via a Convolutional Neural Network, metric learning via Prototypical Networks, frozen out-of-species transfer via a truncated wav2vec 2.0 (layer-2) embedding with a lightweight linear head, and unsupervised anomaly detection via a Convolutional Autoencoder with a One-Class SVM baseline. All five models are trained on a seven-class acoustic dataset of 8,648 clips assembled from ESC-50, AudioSet, UrbanSound8K, Mozilla Common Voice, and Freefield1010, and evaluated on identical hardware under a common nine-technique framework targeting AUC-ROC above 0.85 and per-class F1 above 0.80. The winning model is deployed with TensorFlow Lite INT8 quantisation and Grad-CAM explainability, delivering GPS-tagged SMS alerts including precise coordinates acquired from the SIM808 via AT command to rangers within 30 seconds at under USD 80 hardware cost. It is hypothesised that the out-of-species embedding paradigm will generalise most reliably across all seven classes, building on Geldenhuys and Niesler (2026), who recently demonstrated that this approach matches supervised performance for elephant vocalisations without any fine-tuning.

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Wildlife conservation across Africa is confronting a structural technological deficit with real and measurable consequences. The African elephant population has declined from approximately 1.3 million individuals in 1979 to around 500,000 today, a reduction of more than 60 percent driven primarily by poaching for ivory (Elephanatics Foundation, 2024). Between 10,000 and 15,000 elephants continue to be killed by poachers every year, and the IUCN now classifies savanna elephants as Endangered and forest elephants as Critically Endangered (IUCN, 2021). The United Nations Office on Drugs and Crime published its third World Wildlife Crime Report in 2024, documenting illegal trade affecting more than 4,000 animal and plant species across 162 countries between 2015 and 2021, and concluding that wildlife trafficking has not been substantially reduced in two decades of concerted international action despite an estimated annual criminal revenue of USD 20 billion (UNODC, 2024).

The operational challenge facing conservation rangers is principally one of spatial coverage. A peer-reviewed study published in Nature Sustainability in 2022 established that the current global ranger coverage density stands at one ranger per 72 km², far below the minimum required for effective protection of most threatened ecosystems (Waldron et al., 2022). African Parks, the largest NGO managing African protected areas, employs 2,119 rangers to protect 20 million hectares across 22 parks, approximately one ranger per 94 km² (African Parks, 2024). Akagera National Park in Rwanda, the primary design reference for this project, spans 1,122 km² and has undergone significant biodiversity recovery following renewed investment in conservation infrastructure; however, its borders and internal zones present coverage challenges consistent with the wider continental pattern. Poaching operations exploit these coverage gaps deliberately and systematically, operating at night, in dense vegetation, and at the known intervals between patrol routes.

Passive Acoustic Monitoring (PAM) offers a genuinely complementary capability to traditional visual monitoring. Sound propagates through dense vegetation and operates independently of lighting conditions, allowing a gunshot, a running chainsaw, or an approaching vehicle engine to be detected hundreds of metres from its source. Contemporary AI-based PAM systems have demonstrated reliable acoustic threat detection where connectivity is available; however, all currently deployed real-time acoustic monitoring systems depend on continuous internet or cellular data connectivity for cloud-based inference connectivity that is unavailable across the majority of remote African national parks (Lynam et al., 2025). This project builds Alertreck: a fully offline edge AI acoustic monitoring system that classifies seven threat categories, identifies the specific threat class with a confidence score, appends GPS device coordinates from the SIM808 receiver, and delivers a labelled, GPS-tagged SMS alert to a ranger's mobile phone within 30 seconds with no internet required at any stage.

1.2 Problem Statement

The core problem is not that rangers lack information, it is that they receive it too late, from too few locations, and only after a threat has already materialised. The closest existing solution, Rainforest Connection (RFCx), uses repurposed Android devices to stream audio to cloud-based convolutional neural network servers for near-real-time classification. Field deployments in Cameroon, Borneo, and the Amazon have produced documented detections of illegal activity (Lynam et al., 2025). However, the RFCx Guardian's own technical documentation specifies a minimum continuous data uplink of 50 Kbps for its core classification function (RFCx, 2023). In environments such as Kafue National Park in Zambia covering 22,000 km² with an estimated 4,000 to 6,000 active poachers operating inside its boundaries, reliable cellular data coverage does not exist across the majority of the park. RFCx is therefore architecturally non-functional in precisely the environments where acoustic monitoring is most urgently needed.

The second problem is that offline alternatives lack operational alert delivery. AudioMoth (Hill et al., 2019), a low-cost open-source recorder priced at approximately USD 50, performs on-device event classification using Hidden Markov Model classifiers without requiring internet connectivity. However, when AudioMoth detects a gunshot or chainsaw, the detection is stored on the device's memory card and remains there until a ranger physically retrieves the device which may occur hours or days after the event. A threat that is detected but not communicated within a response window of minutes provides no operational value. Furthermore, AudioMoth relies on pre-programmed acoustic templates that limit generalisation to novel threat sounds and provide no classification of what type of threat was detected.

The third problem is the absence of any system that combines offline classification with explainable alerts and labelled location context. Existing systems either classify without location (AudioMoth, RFCx) or provide spatial information only through commercial infrastructure requiring significant cost and connectivity (ShotSpotter). Rangers who receive automated alerts from black-box systems frequently distrust them, contributing to alarm fatigue and reduced response rates (Lynam et al., 2025; Campos et al., 2019). Alertreck addresses all three gaps simultaneously: it operates entirely offline, delivers real-time GPS-tagged SMS alerts within 30 seconds via GSM cellular, and provides Grad-CAM spectrogram overlays that show rangers exactly which acoustic feature triggered each classification at a hardware cost under USD 80.

A fourth and methodologically critical problem is the absence of empirical guidance on which machine learning paradigm best serves anti-poaching acoustic detection under the dual constraints of limited labelled African data and severe domain shift between training corpora and field deployment. Annotated bioacoustic data are notoriously scarce: training a fully supervised deep network on African gunshot or chainsaw recordings is impractical because such recordings do not exist at the scale supervised learning requires. Geldenhuys and Niesler (2026) recently established that pretrained acoustic embeddings classify elephant vocalisations at performance approaching that of end-to-end supervised neural networks without any fine-tuning of the embedding model and that the second layer of wav2vec 2.0 retains sufficient information for effective elephant call classification while preserving only approximately 10 percent of the full network's parameters. Their result was demonstrated on bioacoustic calls only. Alertreck extends this finding to a multi-class taxonomy that includes non-biological threats (gunshots, chainsaws, vehicles) and benchmarks it directly against supervised, metric-learning, and unsupervised paradigms on identical edge hardware.

1.3 Project's Main Objective

This project aims to design, develop, and evaluate **Alertreck,** a low-cost, fully offline edge AI acoustic monitoring system for remote African conservation environments where no internet connectivity is available.

The core research conducts a formal four-paradigm comparative study, benchmarking supervised classification, metric learning, frozen out-of-species transfer, and unsupervised anomaly detection to determine which learning paradigm most reliably identifies specific poaching-related acoustic threats on identical edge hardware.

Beyond model selection, the project integrates SIM808 GPS coordinate tagging and Grad-CAM explainability into the deployed system, evaluating whether GPS-tagged SMS alerts including precise device coordinates can be consistently delivered to rangers within the 30-second operational response constraint.

1.3.1 List of Specific Objectives

1. To review academic and applied literature on AI in conservation, passive acoustic monitoring, metric learning, frozen out-of-species transfer with self-supervised embeddings, and compound acoustic augmentation; to assemble and standardise a seven-class acoustic dataset of at least 8,000 clips from publicly available sources; and to produce identical train, validation, and test splits for all five model architectures, ensuring a reproducible experimental baseline.
2. To develop and implement Alertreck's five-model comparative architecture comprising a Convolutional Neural Network, a Prototypical Network, a truncated wav2vec 2.0 (layer-2) frozen embedding with a lightweight linear classifier head, a Convolutional Autoencoder, and a One-Class SVM across four learning paradigms; and to integrate the winning model with TensorFlow Lite INT8 quantisation, Grad-CAM explainability, SIM808 GSM/GPS alert delivery with device coordinates appended to every SMS, and a Flask LAN dashboard on a Raspberry Pi 4 (2 GB RAM) prototype assembly.
3. To evaluate system performance across nine measurable techniques targeting AUC-ROC above 0.85, per-class F1 score above 0.80, false-positive rate below 20 percent, end-to-end SMS alert latency under 30 seconds, continuous power consumption at or below 5 watts, GPS coordinates successfully appended to every alert via SIM808 AT+CGNSINF, and prototype hardware cost at or below USD 80; and to produce a formal four-paradigm comparative results table providing reproducible empirical guidance for conservation deployment decisions.

1.4 Research Questions

RQ1. Which learning paradigm supervised multi-class classification, metric learning via prototypical networks, frozen out-of-species transfer, or unsupervised anomaly detection achieves the highest AUC-ROC, per-class F1 score, and lowest false-positive rate when classifying seven-class African savannah acoustic threat data on identical edge hardware?

RQ2. How does per-class detection performance differ across the five models at signal-to-noise ratios of 5, 10, and 20 dB, and which sound class poses the greatest detection challenge under field-representative interference conditions?

RQ3. Can the SIM808 GSM/GPS module acquire a GPS fix and successfully append device coordinates to a labelled SMS alert within the 30-second end-to-end latency constraint, and what is the typical GPS fix acquisition time under outdoor field conditions?

RQ4. Do Grad-CAM heatmaps applied to the winning model's mel-spectrogram inputs highlight time-frequency regions that correspond to the acoustically expected features of each threat class, providing interpretable explanations that support conservation ranger trust in the system?

RQ5. Does the out-of-species embedding advantage demonstrated by Geldenhuys and Niesler (2026) for biological vocalisations specifically, that a truncated, frozen wav2vec 2.0 layer-2 embedding approaches end-to-end supervised performance for elephant calls extend to non-biological acoustic threats (gunshots, chainsaws, and vehicles) in a multi-class African savannah dataset, and does it retain that advantage on edge hardware after INT8 quantisation?

1.5 Project Scope

Alertreck encompasses the design, development, prototype assembly, and performance evaluation of an edge AI acoustic monitoring system for remote African savannah conservation environments, tested using a controlled acoustic testbed replicating the infrastructure constraints of deployment parks. The acoustic taxonomy covers seven sound classes: background_animals (non-threat wildlife), background_wind_rain (weather and natural ambience), threat_gunshot, threat_chainsaw, threat_vehicle, threat_human, and threat_dog. Akagera National Park in Rwanda serves as the primary design reference. Kafue National Park in Zambia, Queen Elizabeth National Park in Uganda, and the Kavango-Zambezi Transfrontier Conservation Area are the primary need-case examples, representing the scale and infrastructure constraints the system is designed to address.

In scope: a seven-class labelled acoustic dataset from publicly available sources; five trained and evaluated model architectures across four learning paradigms; one Raspberry Pi 4 prototype (2 GB RAM) with USB microphone and SIM808 GSM/GPS module; GPS coordinate tagging verified under outdoor conditions; Grad-CAM explainability on the winning model; a Flask LAN dashboard; and a formal nine-technique comparative evaluation study. Explicitly out of scope: multi-species ecological monitoring beyond the seven-class taxonomy; drone or satellite integration; cloud-connected architectures; deployment in an actual national park during the project period; full position triangulation across multiple distributed devices; and large-scale hardware rollout beyond the single prototype unit.

1.6 Significance and Justification

Alertreck is significant for a reason that is easy to state and harder to find in most conservation technology research: it is designed to actually work in the places it is meant to be used. The offline-first architecture, the GSM-only communication pathway, and the sub-USD-80 hardware budget are not design compromises; they are the fundamental design premises, derived directly from the infrastructure realities of remote African national parks. Every technical decision in the system is constrained by the question: will this function without an internet connection, in an outdoor African environment?

The project makes four distinct contributions to the field. To the best of the author's knowledge, no prior work has conducted a formal four-paradigm comparative study of supervised classification, metric learning, frozen out-of-species transfer, and unsupervised anomaly detection for African savannah poaching threat sounds on identical data and hardware, providing reproducible empirical guidance for conservation deployment decisions that the field currently lacks. Additionally, the application of prototypical networks to anti-poaching acoustic classification represents an underexplored direction in the literature, directly addressing the documented background/threat spectral confusion problem by optimising explicitly for inter-class separation in the embedding space. Furthermore, the project extends the Geldenhuys and Niesler (2026) out-of-species embedding finding previously demonstrated for bioacoustic vocalisations into a multi-class taxonomy that includes non-biological threats such as gunshots, chainsaws, and vehicles, testing whether the layer-2 wav2vec 2.0 transfer advantage holds when the target domain encompasses both biological and mechanical sounds rather than a single species.

Fourth, the integration of the SIM808's built-in GPS receiver with offline multi-class acoustic classification transforms point-of-detection alerts into GPS-tagged notifications that tell rangers both what was detected and precisely where the sensor that detected it is located at sub-USD-80 total hardware cost. Grad-CAM explainability for labelled conservation alerts addresses the ranger trust gap identified by Lynam et al. (2025) at an operational depth no existing deployed system provides: rangers can see exactly which acoustic feature in the spectrogram triggered each alert. Beyond the immediate application, the methodology is directly replicable to other keystone species, other ecosystems, and other behavioural monitoring challenges where the signal of interest is deviation from a learned normal.

1.7 Research Budget

The table below presents the hardware bill of materials for the Alertreck prototype unit. All components are commercially available from local electronics suppliers in Rwanda or via regional distributors. The prototype targets a total material cost at or below USD 80 for a minimal viable unit.

Component	Specification	Role	Cost (USD)
Raspberry Pi 4 Model B	2 GB RAM, quad-core ARM Cortex-A72 @ 1.8 GHz	Edge processing audio capture, inference, alert transmission	$53
USB microphone	Plug-and-play, 100 Hz–16 kHz frequency response	Continuous single-channel outdoor audio capture at 44.1 kHz	$8
SIM808 GSM/GPS module	Quad-band 850/900/1800/1900 MHz; built-in GPS receiver; UART serial	Dual-function: GSM for SMS alert delivery + GPS for device coordinates appended to every alert via AT+CGNSINF	$22

Component	Specification	Role	Cost (USD)
SIM card (MTN/Airtel)	Local Rwandan prepaid SIM	Cellular network access for SMS delivery; no data plan required	$2
MicroSD card	32 GB minimum, Class 10	OS, model weights, audio logs, SQLite database	$5
Miscellaneous	UART serial cable, USB-A, jumper wires	Hardware assembly and connectivity	$5
TOTAL			USD 95

Table 9: Hardware Bill of Materials and Research Budget

1.8 Research Timeline

The project is implemented across ten weeks following a Design and Development Research (DDR) methodology. The Gantt chart below presents each phase, its weekly span, and the key deliverables produced.

Phase / Deliverables	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10 +
Phase 1: Data & Preprocessing	█	█								
Phase 2: Supervised & Metric Models			█	█	█					
Phase 3: Frozen transfer and unsupervised detectors (W2V2-L2, Conv-AE, OC-SVM)					█	█				
Phase 4: Comparative Evaluation							█			

Phase 5: Edge Deployment								■		
Phase 6: Direction + GSM Integration								■	■	
Phase 7: Grad-CAM, Dashboard & Evaluation									■	■
Phase 8: Writing & Submission										■

Table 11: Ten-Week Implementation Timeline

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews academic and applied literature relevant to the Alertreck project across seven principal domains: strategic AI in conservation; methodological foundations of Passive Acoustic Monitoring; metric learning and prototypical networks for audio; audio transformer architectures; frozen out-of-species transfer with self-supervised speech embeddings; anti-poaching acoustic monitoring in African ecosystems; and compound acoustic augmentation with curriculum training for robust classification. The search targeted peer-reviewed articles, conference proceedings, and technical reports published between 2015 and 2026 on platforms including Google Scholar, IEEE Xplore, the ACM Digital Library, Frontiers, MDPI Sustainability, arXiv, and the DCASE Challenge proceedings. Approximately forty primary sources were identified and reviewed, of which twenty-eight are directly cited in the proposal. The purpose is not to catalogue every relevant paper but to synthesise the most directly applicable work, compare methodologies and deployment constraints, and identify precisely where the proposed research extends beyond what has been published. The review is structured to map directly onto the five theoretical pillars of the Alertreck conceptual framework: bioacoustics, machine learning across four paradigms, edge computing, explainable AI, and acoustic signal processing.

2.2 Overview of Existing Systems

Three deployed systems represent the current state of the art in real-time conservation acoustic monitoring. The Rainforest Connection (RFCx) Guardian uses repurposed Android devices to stream continuous audio to cloud-based convolutional neural network servers, where gunshots and chainsaw sounds are classified in near real time. Field deployments in Cameroon, Borneo, and the Amazon have produced documented detections of illegal logging and poaching activity (Lynam et al., 2025). However, the Guardian's technical documentation specifies a minimum continuous data uplink of 50 Kbps for its core classification function (RFCx, 2023). Without this uplink, the device records audio but cannot classify it; no alert is ever produced. In Kafue National Park in Zambia, reliable cellular data coverage does not exist across the majority of the park's 22,000 km². RFCx is architecturally non-functional in precisely the environments where acoustic monitoring is most urgently needed.

The AudioMoth (Hill et al., 2019) is a low-cost open-source recorder priced at approximately USD 50 that performs on-device event classification using pre-programmed Hidden Markov Model classifiers, without internet connectivity. Unlike RFCx, all processing runs locally on the device. However, AudioMoth has a critical operational limitation: it has no alert delivery mechanism. When AudioMoth detects a gunshot or chainsaw, the detection is stored on the device's memory card and remains there until a ranger physically retrieves the device which may occur hours or days after the event. Furthermore, AudioMoth relies on pre-programmed acoustic templates, limiting generalisation to novel threat sounds and providing no threat class specificity beyond a binary trigger.

The ARBIMON platform enables large-scale remote audio collection through centralised web-based processing infrastructure, designed primarily for ecological analysis rather than operational threat alerting. Like RFCx, it requires continuous connectivity for core processing and provides no real-time ranger alert delivery. EarthRanger aggregates data from multiple sensor types for conservation managers but is not itself a detection system; it depends on external sensors to provide inputs. Camera traps offer a complementary visual detection capability but require line of sight, adequate lighting, and are rendered ineffective by vegetation density and night-time operations.

Together, these systems define the gap that Alertreck occupies. No currently deployed system simultaneously provides offline acoustic threat classification with specific class labelling, GPS-tagged SMS alert delivery with device coordinates, and Grad-CAM explainability to rangers at sub-USD-80 hardware cost.

2.3 Review of Related Work

2.3.1 AI in Conservation: Strategic Imperatives

Kumar and Jakhar (2022) document how AI addresses the manpower shortages, safety risks, and physical constraints inherent in traditional conservation management by enabling continuous automated surveillance in conditions incompatible with sustained human presence. Isabelle and Westerlund (2022), reviewing AI deployments within the SDG 15 framework, catalogue the

PAWS (Protection Assistant for Wildlife Security) system in Uganda's Queen Elizabeth National Park, which uses game-theoretic patrol route optimisation to predict poaching locations and direct rangers accordingly, with documented reductions in poaching incidents. A persistent bias in conservation AI research, however, favours visual monitoring approaches. Fergus et al. (2024) evaluate CNN and vision transformer platforms for thermal and optical species detection across multiple continents, noting that they share a fundamental constraint: they require line of sight, are rendered ineffective by vegetation density, and cannot detect acoustic threats. Lynam et al. (2025) identify three specific barriers to widespread PAM deployment: bandwidth requirements for cloud uplinks, power consumption constraints, and ranger trust gaps, all three of which are directly addressed by the Alertreck design.

2.3.2 Methodological Foundations of Passive Acoustic Monitoring

Gibb et al. (2018) provide the canonical methodological reference for PAM system design, reviewing the pipeline from audio acquisition through spectral analysis to classification, and documenting practical trade-offs between sampling rate, bit depth, storage capacity, battery life, and detection accuracy. Their analysis is directly relevant to the Alertreck sampling rate choice: 44.1 kHz is required to capture the full harmonic content of chainsaws and vehicle engines and to provide sufficient temporal resolution for SIM808 GPS coordinate tagging. Huzaifah (2017) demonstrated that log-mel spectrograms consistently outperform raw MFCCs as inputs to convolutional classifiers on environmental sound benchmarks, because they preserve more spatial structure in the time-frequency domain. Alertreck therefore uses 128-bin log-mel spectrograms for CNN-based models and 40-coefficient MFCCs for the One-Class SVM, matching each model's input representation to its architectural requirements. Sharma et al. (2023), synthesising 54 AI-based acoustic monitoring studies, confirm that CNNs applied to log-mel spectrograms achieve the current accuracy benchmark for wildlife sound classification, but identify CNNs' systematic overfitting to majority classes on imbalanced datasets as a critical constraint. Alertreck addresses this through focal loss (Lin et al., 2017), which down-weights easy examples and forces the model to focus on difficult minority threat classes.

2.3.3 Metric Learning and Prototypical Networks for Audio

Metric learning for audio classification begins from a different premise than standard supervised learning: rather than learning fixed class boundaries, it learns an embedding function that maps audio inputs to a representation space where semantically similar sounds are geometrically close and dissimilar sounds are geometrically distant. Snell, Swersky, and Zemel (2017) introduced the prototypical network as the canonical implementation, representing each class by a prototype vector computed as the mean embedding of its support examples. The prototypical loss function optimises explicitly for inter-class separation, directly addressing the spectral confusion problem between background and threat acoustic classes. Wang (2020) applied prototypical networks to the DCASE bioacoustic few-shot bird recognition challenge, demonstrating improvements in F-score of 15 to 25 percent over CNN softmax baselines when evaluated on out-of-distribution recordings precisely the conditions that mirror Alertreck's deployment scenario. Nolasco et al. (2023) confirmed that metric learning systems consistently outperformed supervised baselines under distributional shift at the DCASE 2023 challenge. The application of prototypical networks to anti-poaching acoustic threat classification has not been published; this is one of the methodological contributions of the Alertreck comparative study.

2.3.4 Audio Transformer Architectures and the Case Against Fine-Tuning Them on Edge

The Audio Spectrogram Transformer (AST) by Gong, Chung, and Glass (2021) adapts the Vision Transformer architecture to audio by treating a log-mel spectrogram as a sequence of patches processed with multi-head self-attention. On AudioSet, AST achieved mean Average Precision of 0.485, surpassing all previously published CNN approaches. Its self-attention mechanism captures long-range temporal and spectral dependencies that convolutional layers with fixed receptive fields cannot model particularly valuable for sounds with time-varying harmonic structure such as chainsaws and vehicle engines. The tiny-AST variant reduces parameter count to approximately 5 million through layer pruning, bringing inference latency within the feasible range for Raspberry Pi hardware after INT8 quantisation. Jacob et al. (2018) demonstrate that INT8 post-training quantisation of transformer architectures reduces model size by approximately 75 percent with less than 2 percent degradation in classification accuracy, consistent with Alertreck's 10 MB model size non-functional requirement.

However, fine-tuning AST on a small target dataset is precisely the failure mode that Geldenhuys and Niesler (2026) warn against: small annotated bioacoustic corpora cause supervised fine-tuning to overfit and to generalise poorly under domain shift. The Alertreck comparative study therefore drops tiny-AST in favour of a better transfer approach a truncated, fully frozen wav2vec 2.0 layer-2 embedding with a lightweight downstream linear head which retains the representational benefits of a large pretrained encoder while requiring no fine-tuning and an order of magnitude fewer trainable parameters. The rationale is developed in the next section.

2.3.5 Frozen Out-of-Species Transfer with Self-Supervised Embeddings

Geldenhuys and Niesler (2026), in "From Birdsong to Rumbles: Classifying Elephant Calls with Out-of-Species Embeddings", establish a result of direct methodological consequence for the Alertreck project. They show that pretrained acoustic embeddings drawn from general audio, speech, and bioacoustic domains, all of which are either out-of-domain (containing no bioacoustic data) or out-of-species (containing no elephant call data) classify elephant vocalisations at performance approaching that of end-to-end supervised neural networks, without any fine-tuning of the embedding model. The embedding networks themselves remain fixed; only lightweight downstream classifiers, including a linear model and several small neural networks, are trained on top of them. Their best-performing system, built on Perch 2.0, achieved cross-validated AUCs of 0.849 on African bush elephant (Loxodonta africana) calls and 0.936 on Asian elephant (Elephas maximus) calls within 2.2 percent of an end-to-end supervised elephant call classification system.

A second and more practically consequential finding emerges from their layerwise analysis of pretrained transformer encoders. The intermediate representations of self-supervised speech models, specifically the second layer of both wav2vec 2.0 and HuBERT encode sufficient information for effective elephant call classification, and truncation at this layer preserves classification performance while retaining only approximately 10 percent of the parameters of the full network. Compact embedding networks of this kind are, in the authors’ own words, "well suited to on-device processing where computational resources are limited" which is exactly the deployment constraint Alertreck operates under. This result motivates the project’s frozen transfer arm: a truncated wav2vec 2.0 (layer-2) encoder with a lightweight linear head, abbreviated W2V2-L2 throughout this proposal.

Geldenhuys and Niesler's evaluation, however, was restricted to elephant calls; their out-of-species claim has not been tested on non-biological threat sounds. Whether the same advantage extends to the mechanical and human acoustic signatures that dominate anti-poaching operations gunshots, chainsaws, vehicles, human voices is an open question. Research Question 5 of this project answers it directly, by including W2V2-L2 as the frozen-transfer arm of a four-paradigm comparison on a seven-class African savannah dataset that mixes biological (animal vocalisations) with non-biological (mechanical, human) classes. This extension is the most direct novel contribution of the present work to the published literature.

2.3.6 Anti-Poaching Acoustic Monitoring in African Ecosystems

Wrege et al. (2017) deployed acoustic sensor arrays across 55 sites in Central Africa, accumulating over 335,000 hours of audio and achieving recall rates of 0.80 for forest elephant rumbles and 0.94 for gunshot events. Their work demonstrates that distributed acoustic monitoring at a meaningful conservation scale is achievable, and that acoustic data can serve as a long-term record for quantifying hunting intensity and evaluating patrol effectiveness. The limitation from an operational alerting perspective is the absence of real-time alert delivery and explanatory information for ranger response. Hill et al. (2019) showed reliable gunshot detection within 500 metres in tropical forest conditions using on-device HMM classifiers at low hardware cost, but without any alert delivery mechanism. Lopez-Tello and Muthukumar (2018) validated SVM-based acoustic classification for wildlife monitoring under significant acoustic interference, establishing SVM approaches as credible baselines. Campos et al. (2019) document that false-positive rate management is both a critical operational concern and false alerts cause ranger alarm fatigue and systematically underserved in the literature, a gap that Alertreck's nine-technique evaluation framework directly addresses through a dedicated held-out false-positive rate test.

2.3.7 Compound Acoustic Augmentation and Curriculum Training

A central limitation of prior acoustic monitoring work is the gap between training data and field reality. Datasets such as ESC-50, UrbanSound8K, and AudioSet contain predominantly clean, isolated recordings, whereas real deployment combines several simultaneous degradations background noise, far-field attenuation, reverberation, microphone coloration, and signal dropout. Models trained on atomic single-condition clips generalise poorly when these effects co-occur.

Xie et al. (2026) address this directly in the Mega-ASR framework, showing that training on **compound scenarios** of 2 to 5 simultaneous effects drawn from seven canonical degradation categories produces over 30 percent relative error reduction against single-condition baselines. They further introduce a unified severity parameter swept across the full difficulty range and a learnability filter that discards samples too corrupted to be recoverable. Their **curriculum strategy** of easy, medium, hard stages prevents the model collapse that occurs when classifiers trained at moderate difficulty are exposed to severely degraded inputs. Alertreck adopts an analogous three-phase SNR curriculum (≥15 dB → 10–15 dB → 5–10 dB) for the same reason.

Several augmentation techniques complement the compound strategy. SpecAugment (Park et al., 2019) and SpecAugment++ (Wang et al., 2021) apply time and frequency masking to log-mel spectrograms, yielding 3.6 to 4.7 percent gains and forcing the model to learn the spectral shape of events rather than their temporal position. FilterAugment (Nam et al., 2022) simulates microphone-to-microphone frequency response variation, and Morocutti et al. (2023) demonstrate that convolving training clips with the deployment device's recorded impulse response combined with Freq-MixStyle achieves state-of-the-art device generalisation. Together these techniques provide Alertreck with the most direct mechanism for closing the domain gap between downloaded training corpora and its USB microphone in the field.

Beyond augmentation, Lugo Girao et al. (2026) present DiffVQE, the first reproducible diffusion-based acoustic echo and noise model, exceeding the discriminative DeepVQE benchmark at lower computational cost. Although Alertreck performs classification rather than waveform restoration, DiffVQE signals that the broader robust-audio field is converging on lightweight architectures suitable for resource-constrained deployment. For the ProtoNet arm specifically, Gazneli et al. (2022) show that adding a supervised contrastive loss term to the standard prototypical loss further sharpens inter-class separation in the embedding space directly relevant to high-variance classes such as `threat_human` and `threat_vehicle`.

2.4 Summary of Reviewed Literature

The table below summarises the fifteen most directly relevant reviewed works and their contribution to the Alertreck design.

Author(s) & Year	Domain	Key Finding	Relevance to Alertreck
Wrege et al. (2017)	Anti-poaching PAM	0.94 recall for gunshots across 55 African sites	Primary empirical precedent; establishes PAM viability at scale
Gibb et al. (2018)	PAM methodology	Trade-offs between sampling rate, storage, and detection accuracy	Justifies 44.1 kHz design choice; battery and storage constraints
Sharma et al. (2023)	AI acoustic monitoring review	CNNs + log-mel spectrograms achieve highest accuracy; class imbalance a key constraint	Motivates focal loss; validates log-mel over MFCC for CNN models
Lynam et al. (2025)	Conservation technology review	Bandwidth, power, and ranger trust are three critical PAM barriers	Defines all three problems that Alertreck's design addresses directly
Snell et al. (2017)	Metric learning	Prototypical networks optimise inter-class separation via prototype loss	Foundation of the ProtoNet comparative arm
Wang (2020)	Bioacoustic few-shot learning	ProtoNet achieves 15–25% F-score gain over CNN on out-of-distribution audio	Motivates prototypical network for distributional shift in field deployment
Gong et al. (2021)	Audio Spectrogram Transformer	AST achieves mAP 0.485 on AudioSet, surpassing all CNN baselines	Establishes transformer-class encoders as state-of-the-art; motivates frozen variant

Author(s) & Year	Domain	Key Finding	Relevance to Alertreck
Geldenhuys & Niesler (2026)	Out-of-species transfer for bioacoustics	Frozen pretrained embeddings classify elephant calls within 2.2% of supervised baselines; layer-2 wav2vec 2.0 truncation retains ~10% of parameters with no accuracy loss	Foundation of the W2V2-L2 frozen-transfer arm; motivates RQ5
Hill et al. (2019)	AudioMoth edge recorder	On-device HMM achieves reliable gunshot detection at USD 50	Primary offline competitor; establishes baseline and no-alert gap
Campos et al. (2019)	False-positive rate in PAM	Alarm fatigue is critically underserved; false positives erode ranger trust	Motivates dedicated false-positive test in evaluation framework
Xie et al. (2026) Mega-ASR	Compound acoustic augmentation	>30% relative error reduction using 54 compound acoustic scenarios across 7 degradation types; curriculum training prevents collapse on hard samples	Directly motivates compound augmentation pipeline and curriculum SNR training schedule
Wang et al. (2021) SpecAugment++	Spectrogram augmentation	Masking in both input space and hidden states yields 3.6–4.7% accuracy gains over strong baselines	Integrated into CNN, W2V2-L2, and ProtoNet training pipelines as an input-space augmentation step

Author(s) & Year	Domain	Key Finding	Relevance to Alertreck
Morocutti et al. (2023)	Device impulse response augmentation	DIR augmentation + Freq-MixStyle achieves SOTA device generalisation	Motivates USB microphone impulse response recording and convolution to close domain gap
Nam et al. (2022) FilterAugment	Frequency response augmentation	Random frequency response curves simulate microphone variation; complementary to SpecAugment	Applied in augmentation pipeline to simulate USB microphone frequency response variation
Lugo Girao et al. (2026) DiffVQE	Diffusion-based acoustic enhancement	First reproducible diffusion AEC model; exceeds DeepVQE on echo + noise while reducing complexity and model size	Confirms convergence of robust-audio research on lightweight architectures; justifies URGENT-style compound-degradation evaluation

Table 6: Summary of Reviewed Literature

2.5 Strengths and Weaknesses of Existing Systems

Existing conservation acoustic monitoring systems share several strengths. Cloud-based systems such as RFCx achieve high classification accuracy through large server-side neural networks and benefit from ongoing model improvement through centralised training. AudioMoth has demonstrated that offline, low-cost hardware deployment is operationally achievable and that on-device HMM classification produces reliable gunshot detection in tropical forest conditions. SVM-based approaches offer computational efficiency and well-understood decision boundaries that can be interpreted without specialist machine learning expertise. PAM as a methodology is non-invasive, functions at night and through vegetation, and generates continuous data records suitable for long-term ecological analysis.

However, these systems share critical weaknesses for the target deployment environment. All real-time cloud-connected systems (RFCx, ARBIMON) are non-functional without internet connectivity, which eliminates their utility in the majority of remote African parks. AudioMoth's absence of any alert delivery mechanism means that detections have no operational value until a ranger physically retrieves the device. No existing deployed system combines offline operation, multi-class specificity, GPS-tagged alert delivery, and Grad-CAM explainability at sub-USD-100 hardware cost. No existing system provides explainability of detection decisions to rangers, contributing to the documented alarm fatigue problem. Finally, no published study has conducted a formal four-paradigm comparison of supervised classification, metric learning, frozen out-of-species transfer, and unsupervised anomaly detection for African savannah poaching threat sounds leaving conservation organisations without empirical guidance on which approach best serves their specific deployment context.

2.6 Research Gap and Conclusion

Eight converging gaps in the reviewed literature directly justify the Alertreck research programme. First, no formal four-paradigm comparison of supervised classification, metric learning, frozen out-of-species transfer, and unsupervised anomaly detection has been conducted for anti-poaching acoustic data on identical hardware. Second, prototypical networks have not been applied to anti-poaching acoustic classification despite their demonstrated advantages under distributional shift in bioacoustic contexts. Third, the out-of-species embedding finding established by Geldenhuys and Niesler (2026) for elephant calls has not been tested on non-biological threat classes gunshots, chainsaws, vehicles, human voices and its extension to a mixed biological-and-mechanical multi-class dataset is, to the author's knowledge, unpublished. Fourth, no deployed low-cost system combines offline acoustic classification with GPS-tagged SMS alerts that include both the threat class and the precise device coordinates. Fifth, ranger trust through explainability is critically underserved in deployed systems. Sixth, the specific combination of offline operation, labelled multi-class classification, GPS-tagged SMS alerts, and Grad-CAM explainability at sub-USD-100 cost is unoccupied in the current solution space. Seventh, false-positive rate management across learning paradigms has not been systematically evaluated in conservation acoustic monitoring research. Eighth, and directly informed by Mega-ASR (Xie et al., 2026), no prior edge-deployed conservation acoustic classifier has applied compound multi-condition augmentation: all existing work trains on atomic single-condition examples that do not represent the compound degradation profiles encountered in African savannah field deployment.

The reviewed literature establishes that AI-powered PAM is technically feasible, ecologically justified, and operationally valuable. CNN architectures deliver high accuracy for labelled sounds. Metric learning generalises better under distributional shift. Frozen out-of-species embeddings approach supervised performance with an order of magnitude fewer trainable parameters and are well suited to on-device deployment (Geldenhuys & Niesler, 2026). SIM808 GPS coordinate tagging is proven in deployed systems. Grad-CAM provides interpretable detection evidence. Compound acoustic augmentation with curriculum training, as validated by Mega-ASR on exactly the datasets Alertreck uses (ESC-50, UrbanSound8K, AudioSet), provides the mechanism for closing the gap between benchmark accuracy and field robustness. Alertreck integrates all of these capabilities in a single offline-first system, and its formal four-paradigm comparative evaluation will provide reproducible empirical guidance that the field currently lacks.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the research design, dataset strategy, system analysis, architecture, component design, development toolchain, and ethical framework that guide the Alertreck implementation and evaluation. Each section maps directly to a specific project objective from Chapter One. The research follows a Design and Development Research (DDR) methodology, the appropriate approach for applied computing projects whose primary deliverable is a functional technological artefact evaluated against empirical performance criteria (Richey & Klein, 2014). The project is structured as a controlled comparative experimental study: five model architectures across four learning paradigms are trained on identical datasets, evaluated on identical hardware, and compared across a common nine-technique evaluation framework. This design produces a research contribution that goes beyond demonstrating a working system to providing reproducible empirical guidance for conservation deployment decisions.

3.2 Research Design

The research proceeds through eight sequential phases. Phase 1 (Weeks 1–2) completes dataset acquisition, applies the preprocessing pipeline, and produces identical training, validation, and test splits for all five models. Phases 2 and 3 (Weeks 3–6) train the five models across the four paradigms. Phase 4 (Week 7) conducts comparative evaluation on the held-out test set and selects the deployment candidate. Phase 5 (Week 8) deploys the winning model to the Raspberry Pi 4 with TFLite INT8 quantisation. Phase 6 (Weeks 8–9) integrates the SIM808 GSM/GPS module: verifies SMS delivery via AT commands, tests GPS fix acquisition time outdoors, and confirms GPS coordinates appear correctly in the SMS alert payload. Phase 7 (Weeks 9–10) applies Grad-CAM explainability, builds the Flask LAN dashboard, and completes the full nine-technique evaluation study. Phase 8 (Week 10+) produces the final report and submission package.

The following experimental controls ensure fairness across all five models: identical preprocessed dataset at 44.1 kHz with 3-second windows and 50 percent overlap; identical 60/20/20 stratified train/validation/test split with fixed random seed 42; identical evaluation hardware (Raspberry Pi 4 Model B, 2 GB RAM); identical nine evaluation techniques applied to all models; and identical SNR injection conditions (5, 10, and 20 dB) for the anomaly detection evaluation. Experiment tracking uses local MLflow, consistent with the offline-first design constraint.

3.3 Dataset Description

The Alertreck training corpus is assembled from five publicly available sources selected for acoustic diversity, licensing compatibility, and coverage of the seven target sound classes. As of 20 April 2026, the corpus stands at 8,648 raw audio clips totalling approximately 9.88 hours of audio. Six of the seven target classes have met or exceeded their pre-augmentation count targets; the remaining gap in threat_chainsaw is being addressed through an ongoing AudioSet download.

Source	Licence	Classes Contributed	Clips	Role
AudioSet	CC BY 4.0	threat_gunshot, threat_vehicle, threat_human, background_animals	~5,000	Primary source for gunshot, vehicle, and human threat classes
ESC-50	CC BY-NC 3.0	All 7 classes	880	Balanced 40-clip coverage across all seven classes
UrbanSound8K	CC BY-NC	threat_dog, threat_vehicle	2,000	Dog bark and vehicle engine sounds
Mozilla Common Voice	CC0	threat_human	~500	English speech for human presence detection
Freefield1010	Mixed CC	background_animals	500	Bird calls and outdoor ambient audio

Table 2: Dataset Sources, Licences, and Acquisition Status

All datasets are standardized through an identical deterministic preprocessing pipeline before any model training begins. The pipeline is implemented in Python 3.10 using librosa 0.10+ and audiomentations, producing reproducible .npz feature shards written to data/processed/. A data manifest CSV records the path, class, source, duration, sample rate, and licence for every clip in the corpus.

Step	**Operation**	**Parameters**	**Justification**
1	Resample	Kaiser-best resampling to mono; target: 44.1 kHz mono	Captures full harmonic content of chainsaws, vehicle engines, and dog barks; provides sufficient spectral resolution (Gibb et al., 2018)
2	Normalise	RMS amplitude normalisation; target: −23 dBFS (EBU R128)	Removes gain variation across dataset sources; prevents loudness shortcuts in learned features
3	Segment	Sliding window extraction; window: 3 s, step: 1.5 s (50% overlap)	Ensures full gunshot transient captured regardless of onset position within window
4a	Log-mel spectrogram	STFT → 128-bin mel filterbank → log; 25 ms window, 10 ms hop, Hann	Input representation for CNN, ProtoNet, Conv-AE (Huzaifah, 2017)
4b	Raw waveform (44.1 kHz)	3-second mono waveform, resampled to 16 kHz only for the W2V2-L2 frozen encoder input	wav2vec 2.0 was pretrained at 16 kHz; matching its expected input is required for the frozen layer-2 transfer (Geldenhuys & Niesler, 2026)

Step	Operation	Parameters	Justification
4c	MFCC+Δ+ΔΔ	DCT → delta → delta-delta; 40 coefficients → 120-dim vector per frame	Input representation for OC-SVM; dimensionality reduction makes RBF kernel tractable
5	USB microphone DIR calibration	Sweep-tone impulse response recorded via USB microphone; convolved with all training clips	Closes device domain gap between downloaded training data and deployment microphone (Morocutti et al., 2023)
6a	SpecAugment	Time masking: 2 masks, max 40 frames; frequency masking: 2 masks, max 20 bins	Forces model to learn full spectral shape of threats (Wang et al., 2021)
6b	Compound augmentation	Randomly sample 2–4 effects from: Gaussian noise (5–20 dB SNR), RIR convolution, low-pass filter (2–8 kHz cutoff), MP3 compression (VBR 2–7), gain ±6 dB, clipping distortion	Simulates real compound field conditions; directly validated by Mega-ASR (Xie et al., 2026)
6c	FilterAugment	Random frequency response curve per clip; magnitude: ±6 dB across random sub-band	Simulates USB microphone frequency response variation (Nam et al., 2022)
6d	Mixup	Beta distribution β(0.4); inter-class and intra-class pairs	Additional regularisation; improves class boundary generalisation (Zhang et al., 2018)
6e	Learnability filter	Discard augmented clips where Conv-AE reconstruction error	Removes unlearnable hard-negative samples that

Ste p	Operation	Parameters	Justification
		exceeds 95th percentile of background distribution	destabilise training (Xie et al., 2026)
7	Split	Stratified 60/20/20 split; seed: 42; test set locked before any augmentation applied	Held-out test set reflects clean baseline distribution; augmentation applied only to training folds

Table 3: Preprocessing and Compound Augmentation Pipeline Summary

Steps 6a through 6e are applied during training only. The validation and held-out test sets receive only the base preprocessing (steps 1–5) to ensure evaluation reflects realistic field conditions without artificially inflating robustness metrics. The compound augmentation in step 6b applies between 2 and 4 effects simultaneously per clip, with each effect drawn independently from the available pool and filtered for physical plausibility; for example, RIR convolution is not combined with a dry clipping effect, as these are acoustically contradictory. This compound strategy is directly motivated by Mega-ASR's finding that models trained on atomic single-condition data fail systematically on the compound conditions that characterise real deployment environments. Step 4b is applied only to the wav2vec 2.0 layer-2 frozen-transfer arm, which requires its native 16 kHz pretraining sample rate as input; all other models receive the 128-bin log-mel representation from step 4a.

3.3.1 Curriculum Training Schedule

Following the progressive difficulty strategy validated in Mega-ASR (Xie et al., 2026), all five models are trained under a three-phase curriculum that expands the severity range of the compound augmentation pipeline progressively across the training schedule. This prevents the encoder from learning shortcuts on clean spectral features that collapse under severe compound degradation, the most common failure mode when models are trained on a uniform mixture of all difficulty levels simultaneously.

Phase	Training Weeks	Augmentation Severity	SNR Range	Rationale
Phase A	Weeks 3–4	Clean clips + mild single-condition augmentation (steps 6a, 6d)	≥15 dB SNR	Builds reliable acoustic feature extraction on clean spectral structure
Phase B	Week 5	Medium compound augmentation: 2–3 effects at moderate severity (steps 6a–6d, k≤0.5)	10–15 dB SNR	Introduces realistic field degradation; prevents overfitting to clean training distribution
Phase C	Week 6	Full compound augmentation pipeline: 2–4 effects at full severity including far-field and device distortion	5–10 dB SNR	Exposes models to hardest compound conditions; learnability filter (step 6e) applied to prevent training collapse

Table 5: Curriculum Training Schedule Across Three Difficulty Phases

The curriculum applies identically to the CNN, W2V2-L2, and ProtoNet training pipelines, with one important caveat for the frozen transfer arm: only the lightweight linear head receives gradient updates across the curriculum. The wav2vec 2.0 layer-2 encoder is held fixed in all phases, consistent with the Geldenhuys and Niesler (2026) finding that fine-tuning the embedding model is unnecessary and degrades generalisation under domain shift. For Conv-AE and OC-SVM, which are trained on background-class audio only, the curriculum restricts Phase A training to clean background clips and progressively introduces compound-degraded background clips in Phases B and C. This ensures the anomaly detector builds a robust model of what normal background audio looks like under compound field conditions, rather than only under clean recording conditions which is the primary cause of elevated false-positive rates when anomaly detectors are deployed in environments noisier than their training data.

3.4 System Architecture

Alertreck integrates four hardware and software layers. The Sensing Layer is a single plug-and-play USB microphone (100 Hz–16 kHz frequency response) connected directly to the Raspberry Pi via USB, capturing continuous single-channel outdoor audio at 44.1 kHz. The Processing Layer is the Raspberry Pi 4 Model B (2 GB RAM, quad-core ARM Cortex-A72 at 1.8 GHz), running a continuously active Python audio processing daemon managed by a systemd service, responsible for audio capture, VAD segmentation, feature extraction, TFLite model inference, and Grad-CAM generation. The Communication Layer delivers labelled, GPS-tagged SMS alerts via the SIM808 GSM/GPS module (quad-band 850/900/1800/1900 MHz) over UART serial with AT command control. The SIM808 serves a dual function: its GSM component transmits the SMS alert over the MTN or Airtel Rwanda cellular network, and its built-in GPS receiver provides the device's precise coordinates (acquired via AT+CGNSINF) which are appended to every alert payload. No internet connection is required at any stage. The Interface Layer is a Flask 3.x LAN dashboard served at port 5000 on the Raspberry Pi's network interface, displaying real-time anomaly scores, class labels, GPS coordinates, Grad-CAM overlays, and alert history.

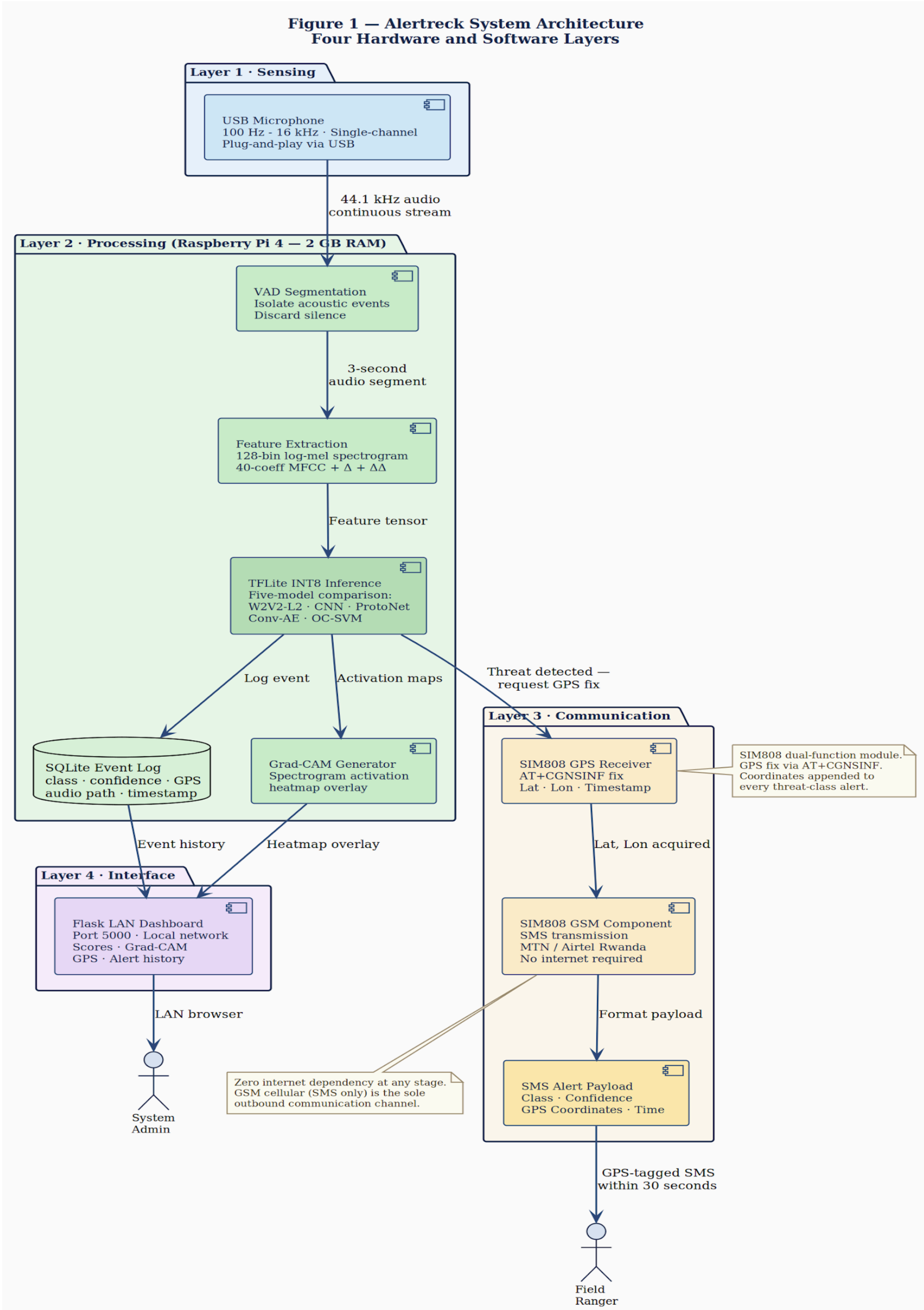

Figure 1: Alertreck System Architecture Four Hardware and Software Layers

3.5 System Design

3.5.1 Five-Model Architecture

Five model architectures are designed across four learning paradigms. All five use identical preprocessed input representations (where compatible) and are trained on identical data splits for fair comparison. The four paradigms supervised classification, metric learning, frozen out-of-species transfer, and unsupervised anomaly detection are each represented by at least one model, with one classical baseline (OC-SVM) included to provide a computational and interpretability floor against which the deep methods are benchmarked.

Model	Paradigm	Input	Training Data	Loss Function	Key Design Choice
CNN	Supervised	128-bin log-mel + SpecAugment	All 7 classes (labelled)	Focal loss γ=2	4 conv blocks + FC; ~2.1M params; encoder reused by ProtoNet; SpecAugment applied at input
ProtoNet (CNN enc)	Metric learning	128-bin log-mel + SpecAugment	All 7 classes (episodic)	Prototypical loss + SupCon (λ=0.1)	N=7, K=5 support, Q=15 query per episode; SupCon term pulls same-class embeddings across episodes (Gazneli et al., 2022); t-SNE visualisation

Model	**Paradigm**	**Input**	**Training Data**	**Loss Function**	**Key Design Choice**
W2V2-L2 + linear head	Frozen transfer (out-of-species)	16 kHz raw waveform → frozen wav2vec 2.0 layer-2 embedding (768-dim mean-pooled)	All 7 classes (labelled); only linear head trained	Cross-entropy + focal weighting	wav2vec 2.0 base model truncated at layer 2 preserves ~10% of full parameters (Geldenhuys & Niesler, 2026); encoder frozen; only a 768→7 linear classifier is trained; no fine-tuning of the embedding
Conv-AE	Unsupervised anomaly	128-bin log-mel + compound aug	Background only (curriculum)	MSE reconstruction	3-layer encoder to 16×16×128 latent; symmetric decoder; curriculum trains on progressively harder compound-degraded background audio
OC-SVM	Classical anomaly baseline	120-dim MFCC+Δ+ΔΔ + compound aug	Background only (curriculum)	One-class hinge	RBF kernel; ν=0.1 contamination; grid search for γ; curriculum trains on progressively

Model	Paradigm	Input	Training Data	Loss Function	Key Design Choice
					harder compound-degraded background audio

Table 4: Five-Model Architecture Comparison Across Four Learning Paradigms

The substitution of W2V2-L2 for the tiny-AST architecture used in earlier project drafts reflects a deliberate methodological repositioning. Earlier drafts assumed that a supervised, fine-tuned transformer would provide the strongest representational baseline; the more recent literature, particularly Geldenhuys and Niesler (2026), establishes the opposite that a frozen, truncated self-supervised speech embedding with a lightweight linear head approaches end-to-end supervised performance on out-of-species bioacoustic classification while requiring approximately an order of magnitude fewer trainable parameters. Replacing tiny-AST with W2V2-L2 has two consequences for the experimental design. First, it introduces a genuinely distinct fourth paradigm (frozen transfer), rather than duplicating supervised learning with a second architecture. Second, it provides the empirical vehicle for answering RQ5 directly: whether the out-of-species advantage demonstrated for biological vocalisations extends to non-biological acoustic threats.

3.5.2 Evaluation Framework

System performance is measured across nine evaluation techniques, each addressing a distinct aspect of operational reliability. AUC-ROC curve (target: above 0.85 for all five models) is the primary headline metric, measuring detection performance independently of threshold selection. The per-class F1 score at the operational threshold (target: above 0.80 for all seven classes on supervised, metric, and frozen-transfer models) measures the balance between alert sensitivity and false-positive rate for each threat category. Threshold ablation sweeps the detection threshold across its full range to identify the optimal operating point for field conditions. The false-positive rate test evaluates the model on held-out normal audio comprising rain, wind, and insect recordings, targeting a rate below 20 percent. The anomaly injection test evaluates recall by mixing labelled gunshot, human voice, and vehicle sounds into normal audio at SNR levels of 5, 10, and 20 dB. The latency benchmark measures end-to-end inference time on the Raspberry Pi 4, targeting below 30 seconds. The energy benchmark measures continuous power draw during operation, targeting at or below 5 watts. The GPS integration test verifies that the SIM808 GPS receiver acquires a fix and successfully appends device coordinates to the SMS alert payload within the 30-second latency budget. Grad-CAM validation provides qualitative confirmation that highlighted spectrogram regions match acoustically expected features of each threat class.

3.5.3 UML and Design Artefacts

The complete system design is documented through standard UML artefacts. The Use Case Diagram (Figure 3) depicts interactions between the field ranger, system administrator, and the automated system, covering anomaly alert reception, audio event review, system configuration, and report generation. The Class Diagram (Figure 4) represents the principal software components AudioCapture, FeatureExtractor, ModelInference, GPSModule, AlertManager, and EventLogger and their associations and method signatures. The Entity Relationship Diagram (Figure 5) defines the local SQLite event database schema, including entities for DetectedEvent, AudioSegment, AlertLog, and SystemStatus. The Sequence Diagram (Figure 6) traces the detection event flow from continuous audio monitoring through TFLite inference, GPS fix acquisition, and final SMS dispatch, demonstrating the under-30-second end-to-end latency target. To preserve readability, each UML diagram is presented on its own dedicated page on the pages that follow, with the wide diagrams rotated to landscape orientation; all four figures are embedded as native SVG so they remain crisp at any zoom level.

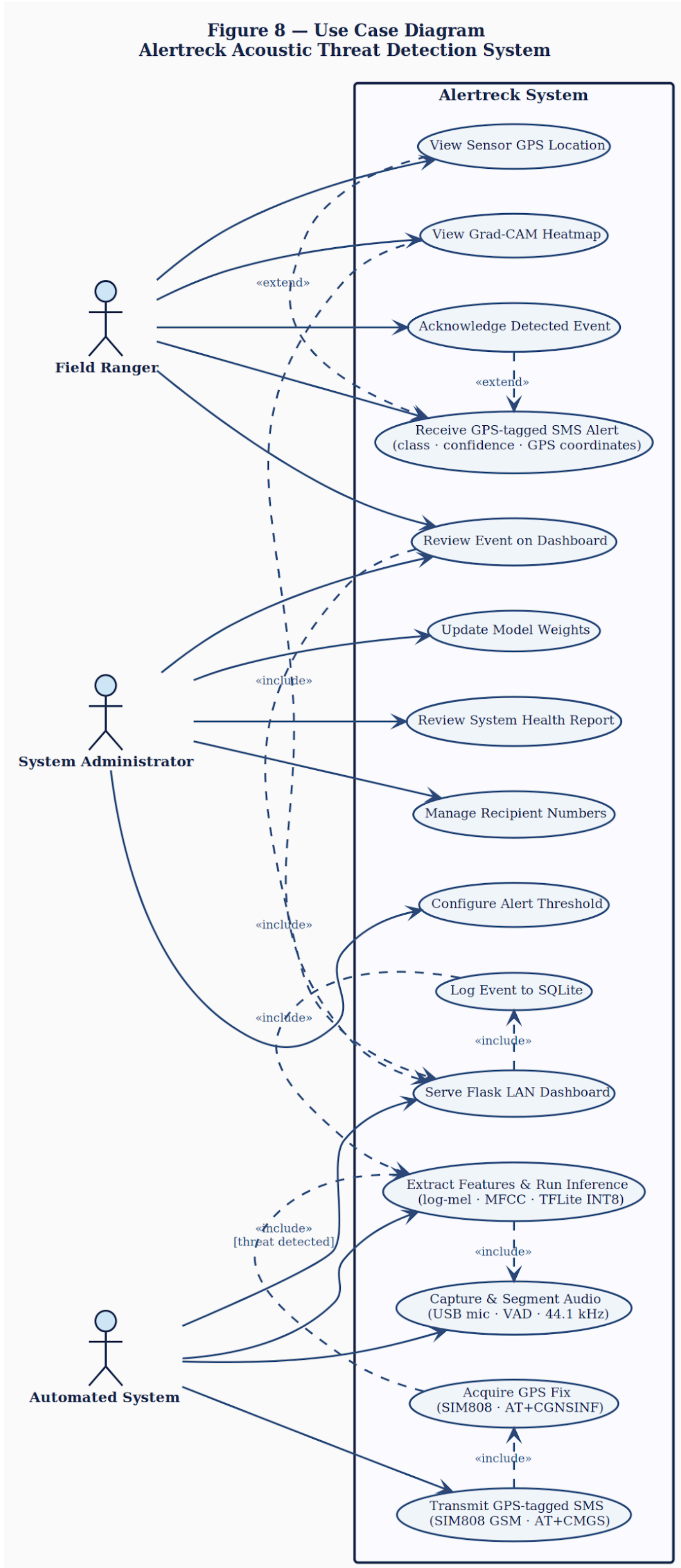

Figure 3: Use Case Diagram Field Ranger, System Administrator, and Automated System

Figure 9 — Class Diagram
Alertreck Software Component Architecture

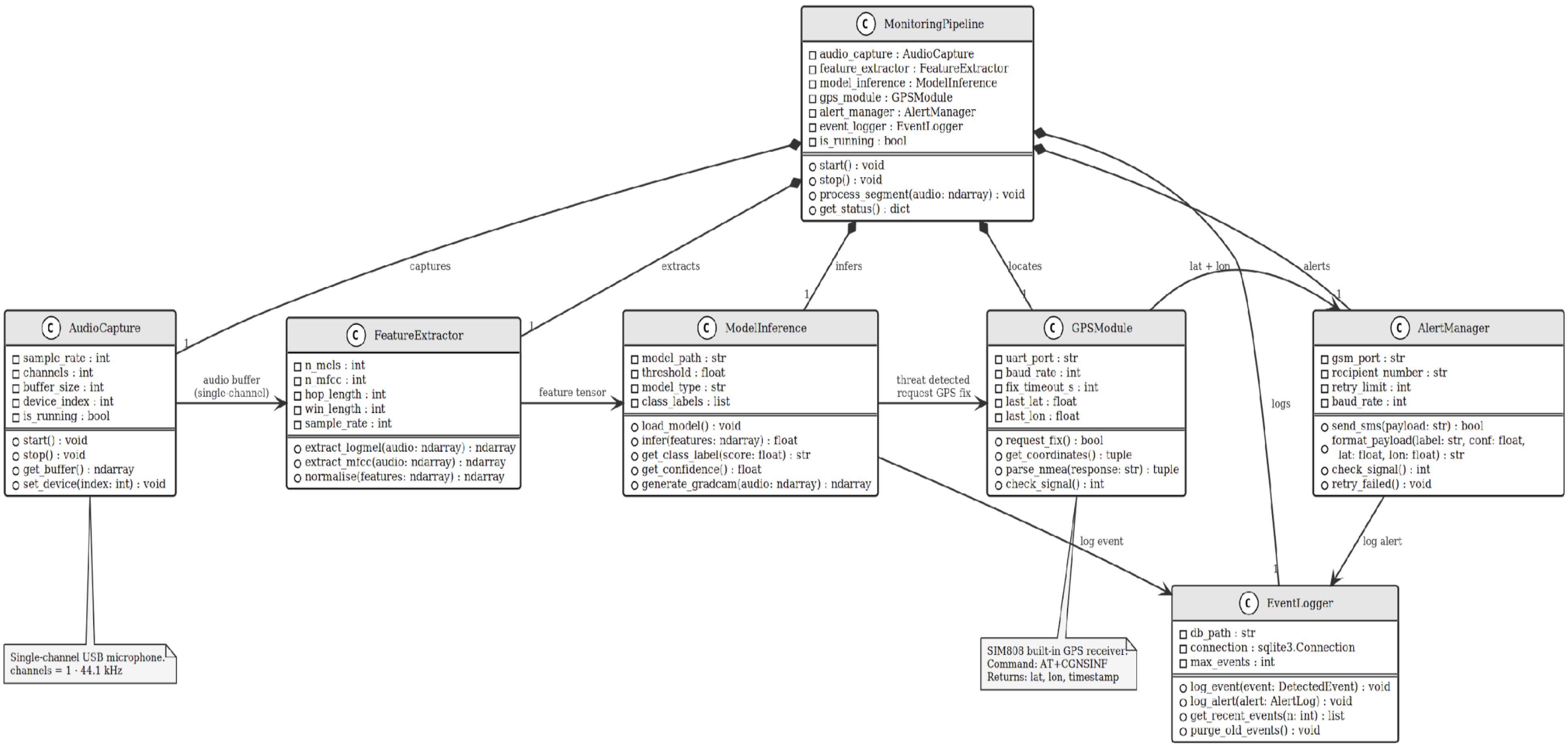

Figure 4: Class Diagram Alertreck Software Component Architecture

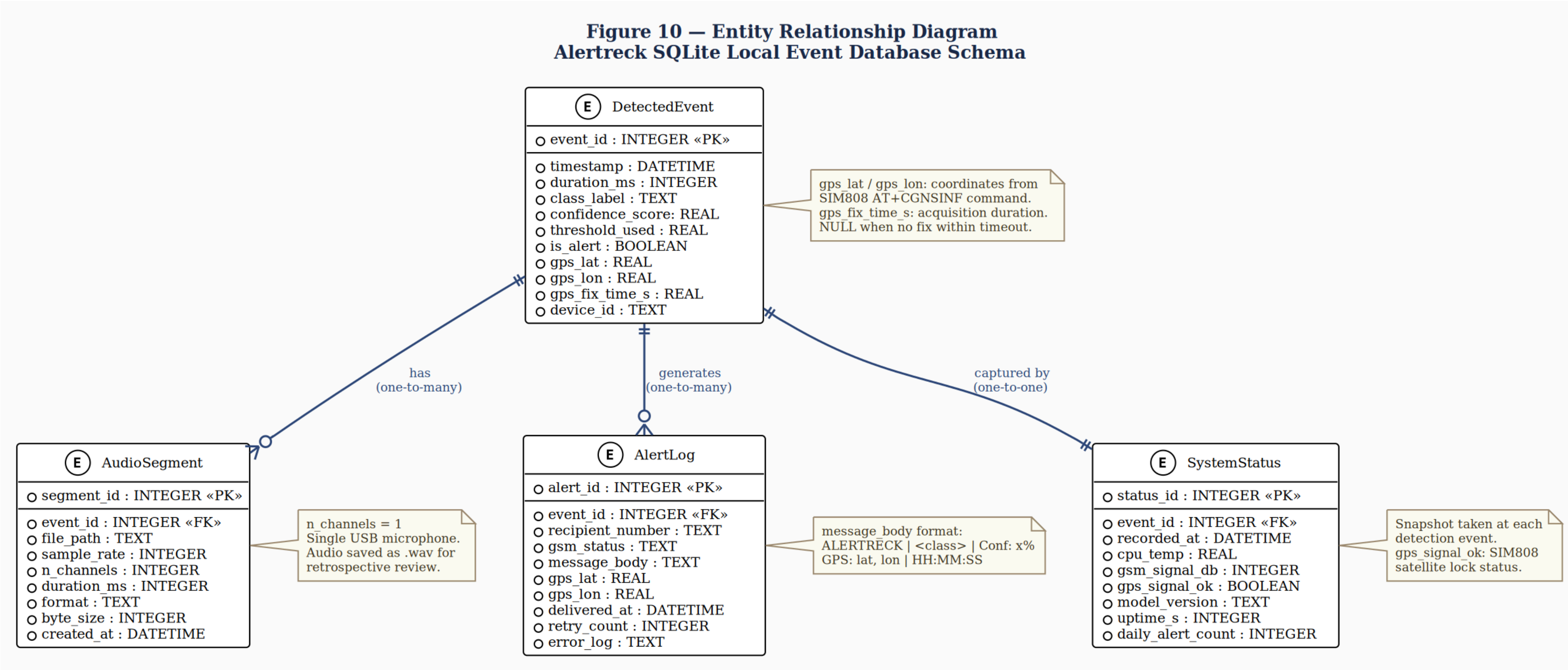

Figure 5: Entity Relationship Diagram Alertreck SQLite Local Event Database Schema

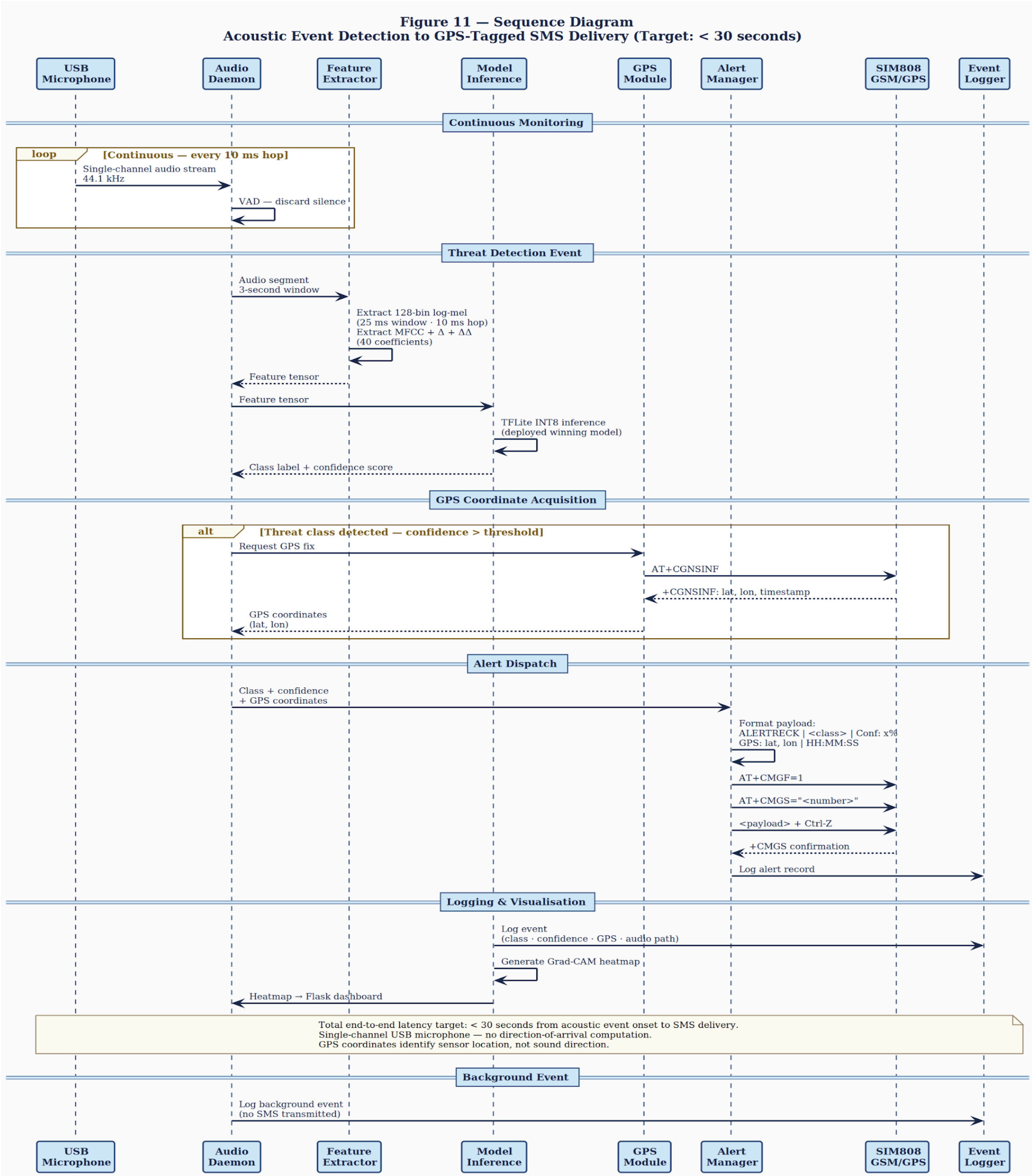

Figure 6: Sequence Diagram Acoustic Event Detection to GPS-Tagged SMS Delivery

3.6 Development Tools

Tool / Technology	Version	Role in Project
Python	3.10	Primary language for all system components, preprocessing, training, and deployment
librosa	0.10+	Audio I/O, resampling, STFT, mel filterbank, MFCC computation
TensorFlow + TensorFlow Lite	2.x	CNN and Convolutional Autoencoder training; INT8 post-training quantisation; Raspberry Pi 4 inference runtime
Hugging Face transformers + torchaudio	4.x / latest	wav2vec 2.0 base model loading; truncation at layer 2 to extract frozen W2V2-L2 embedding; downstream linear head training
PyTorch	latest	Prototypical Network training, episodic data loader, embedding distance computation; supports torchaudio for waveform processing
scikit-learn	1.3+	One-Class SVM training and grid search; confusion matrix; AUC-ROC; t-SNE projection; linear head for W2V2-L2 (LogisticRegression fallback)
audiomentations	latest	Time-shift, pitch perturbation, gain variation, additive noise augmentation pipeline
Flask	3.x	LAN web dashboard: class, confidence, GPS coordinates, Grad-CAM overlay, alert history, audio playback
SQLite	built-in	Local event log database: timestamps, class labels, confidence, GPS coordinates, audio paths

Tool / Technology	Version	Role in Project
pyserial	3.5+	SIM808 GSM/GPS UART serial communication; AT command sequences for SMS transmission
tf-explain / pytorch-grad-cam	latest	Grad-CAM spectrogram activation heatmap generation for the winning model
MLflow	latest (local)	Offline experiment tracking: hyperparameters, metrics, model versions across all five models
Git + GitHub	latest	Version control, commit history, artefact documentation throughout development
Raspberry Pi 4 Model B	2 GB RAM	Edge processing unit: audio capture, VAD, feature extraction, TFLite inference, Grad-CAM, GPS-tagged alert transmission
USB microphone	plug-and-play USB	Single-channel audio capture at 44.1 kHz; plug-and-play USB connection to Raspberry Pi

Table 10: Development Tools and Technologies

3.7 Ethical Considerations

All acoustic data collection is entirely passive and non-invasive. No physical interaction with, capture of, restraint of, or deliberate stimulation of any animal will occur at any stage of the project. Sensor hardware installed in outdoor environments will be positioned to minimise disturbance to natural animal behaviour and habitat structure, using camouflaged enclosures attached to existing structures or trees with non-invasive mounting. Any field data collection activities will require written approval from the relevant national conservation authority and from the African Leadership University institutional research ethics committee before commencement.

Audio data collected during testing will be used exclusively for the stated research purposes and stored securely on encrypted drives accessible only to the principal investigator and named supervisor. No personally identifiable information relating to rangers, community members, or any other individuals will be collected, stored, or processed. All dataset sources are used in compliance with their applicable licences: ESC-50 (CC BY-NC 3.0), UrbanSound8K (CC BY-NC), Freefield1010 (mixed CC Freesound), AudioSet metadata (CC BY 4.0), and Mozilla Common Voice (CC0 public domain). The pretrained wav2vec 2.0 model used for the W2V2-L2 frozen transfer arm is released under the MIT License by its original authors, permitting both research and downstream use. SMS messages transmitted via the SIM808 GSM/GPS module are sent in plaintext over the cellular network, which is a known property of the GSM SMS protocol. The alert payload contains the event class, confidence score, GPS coordinates of the device, and timestamp. While GPS coordinates identify the sensor location, this information is already known to the conservation organisation that deployed the unit. SMS encryption using a pre-shared key is documented as a future extension.

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